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LGCF-Net: Local–Global Cross-Modal Fusion With Residual-Gated Attention for Hyperspectral–LiDAR Land-Cover Classification

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***Abstract*—Hyperspectral imagery (HSI) and LiDAR data provide complementary spectral and structural information, which, when fused, can significantly improve land-cover classification. However, the heterogeneity of these modalities makes feature interaction and redundancy suppression challenging. In particular, existing approaches often overlook the importance of modelling interactions between local and global representations across modalities. This paper proposes LGCF-Net, a hybrid fusion framework that explicitly models local–global feature interactions for HSI–LiDAR classification. The framework consists of four modules. First, the Spectral–Spatial Transition Refinement (SSTR) module extracts hierarchical spectral–spatial features using a pyramid convolution strategy. Second, the LiDAR Feature Enhancement Module (LFEM) extracts spatial features from the LiDAR modality. Third, the Cross-Modal Convolutional Transformer (CMCT) incorporates multiscale convolutional layers into the transformer architecture to capture local–global contextual features, followed by residual-gated cross-modal attention (RGCA) for cross-modal interaction. RGCA facilitates information exchange between HSI and LiDAR while suppressing redundant features. Finally, the Classification Token Mixer (CTM) module aggregates refined features for the final prediction. Extensive experiments on four benchmark HSI–LiDAR fusion datasets demonstrate that LGCF-Net achieves the best OA on MUUFL, Trento, and Augsburg, while attaining the highest AA and Kappa on Houston.**

***Index Terms*—Land-cover classification, multimodal data, transformers, cross attention fusion, HSI, LiDAR.**

I. Introduction

RECENT advances in sensor technology, data acquisition platforms, and computational methods have substantially increased the resolution, diversity, and availability of remote sensing data, including hyperspectral imagery (HSI), LiDAR, and data captured by sensing systems in other modalities. These developments have facilitated a wide range of applications, including land-cover classification [1], [2], agricultural monitoring [3], geological resource exploration [4], and disaster management [5]. Among these, land-cover classification plays a pivotal role in sustainable development, environmental protection, and resource management [6]. Furthermore, these advancements have facilitated the integration of modalities with complementary information, thereby overcoming the spectral ambiguity and spatial insufficiency inherent in single-source data [7]. Recent studies have also explored cross-modal alignment and progressive fusion for heterogeneous remote sensing modalities, including SAR–optical image registration and fusion [8], [9], further highlighting the importance of effective interaction between complementary sensing sources.

For land-cover classification, HSI and LiDAR have emerged as two of the most widely adopted complementary modalities, owing to their distinct but synergistic capabilities [10]. HSI provides rich spectral information that enables the discrimination of surface materials through subtle spectral variations [11], while LiDAR captures the three-dimensional structural geometry of the terrain, providing precise spatial and elevation cues [12]. HSI spectral reflectance and LiDAR-derived structural elevation information provide complementary cues for land-cover classification. Their joint use is particularly beneficial for distinguishing classes that are spectrally similar but structurally different [13]. However, reliable joint classification remains challenging due to the heterogeneous characteristics of these modalities. Differences in spatial resolution, sensing mechanisms, and spectral characteristics between HSI and LiDAR create a modality gap that complicates the learning of unified representations. Bridging this gap is essential for developing robust multimodal fusion frameworks for accurate and generalizable land-cover classification.

Traditional approaches, including shallow machine learning and hand-crafted feature-based methods, have provided a foundation for early research in land-cover classification [14]–[17]. Nevertheless, their limited representational capacity restricts the modeling of complex spectral–spatial relationships inherent in HSI–LiDAR data [18]. Convolutional neural networks (CNNs) have largely surpassed these traditional methods owing to their powerful hierarchical feature extraction capability. Despite their success, CNN-based models remain constrained by their fixed receptive fields and a predominant focus on local spatial context [2]. As HSI data exhibits high spectral dimensionality, global contextual modeling becomes crucial to capture long-range spectral dependencies and inter-band correlations [19]. To overcome this limitation, transformer-based architectures have been introduced [20]–[22], leveraging self-attention to capture long-range dependencies and

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global contextual relationships. However, while transformers excel at modeling global interactions, they tend to overlook fine-grained local structures crucial for detailed land-cover discrimination [23], [24]. Hybrid CNN–transformer models have been proposed to address this challenge [25], [22], [26] yet many existing frameworks still lack effective interaction between local and global representations, limiting their ability to comprehensively capture the spectral–spatial diversity of multimodal data. Moreover, existing multimodal fusion frameworks often lack explicit mechanisms to regulate cross-modal information exchange, which can result in redundant or inconsistent feature integration when modalities exhibit strong heterogeneity. Addressing these limitations requires a unified framework capable of simultaneously modelling local–global intra-modal relationships and controlled inter-modal feature interactions.

Another key challenge arises at the modality level. Given the fundamentally different characteristics of HSI and LiDAR, modality-specific feature extraction is essential; however, most existing methods similarly treat both modalities, neglecting their inherent differences. Moreover, effective multimodal fusion remains a significant challenge, requiring frameworks that harness complementary cross-modal information while minimising redundancy and suppressing irrelevant features [7], [27]. Among fusion strategies, late-fusion methods extract features independently and combine them only at the final stage, which often limits cross-modal interactions and constrains joint representation learning [28]. In contrast, progressive fusion strategies encourage semantic alignment and enable complementary representation learning [29] [25]. Within progressive fusion frameworks, cross-attention is a widely adopted mechanism [25], [30], [31]. However, standard cross-attention modules often lack explicit control over the extent of information exchange between modalities. This limitation can result in unstable or inconsistent feature integration, particularly when the modalities are highly heterogeneous. Therefore, it is essential to develop a unified framework with controlled information exchange during learning. Such integration is key to producing semantically consistent and highly discriminative multimodal representations.

Motivated by these challenges, we propose LGCF-Net (Local–Global Cross-modal Fusion Network), a unified framework for HSI–LiDAR classification that explicitly addresses two key limitations of existing methods: the inadequate joint modeling of local and global representations, and the insufficient regulation of cross-modal feature interaction. Specifically, LGCF-Net integrates multi-scale convolutional operations within the transformer architecture to simultaneously learn local fine-grained structures and global contextual dependencies. Moreover, a Residual-Gated Cross-Modal Attention (RGCA) mechanism is developed to adaptively control information exchange between HSI and LiDAR, thereby reducing redundant feature fusion while preserving complementary modality-specific features.

The main contributions of this work are summarized as follows:

- We propose LGCF-Net, a unified cross-modal fusion framework that simultaneously models intra-modal local–global dependencies and inter-modal HSI–LiDAR interactions within a single optimisation objective, enabling unified and discriminative multimodal representation learning.
- We introduce a spectral–spatial pyramid convolutional attention module to enhance HSI feature representation by capturing multi-scale spectral–spatial patterns while reducing spectral redundancy.
- We develop a Cross-Modal Convolutional Transformer (CMCT) with residual-gated cross-modal attention to jointly model local fine-grained and global contextual features, while adaptively regulating HSI–LiDAR interaction to enhance complementary information and suppress redundant cross-modal feature fusion.
- Extensive experiments on four benchmark HSI–LiDAR datasets demonstrate consistent improvements over state-of-the-art multimodal fusion methods.

II. Related Work

Recently, deep learning techniques have significantly advanced HSI–LiDAR fusion for remote sensing image analysis, enabling improved representation learning and classification performance.

A. CNN-Based HSI–LiDAR Fusion Methods

Prominently, CNN-based methods have been employed for multisource feature extraction and fusion. Most existing methods adopt dual-branch architectures to extract modality-specific features from HSI and LiDAR before performing feature fusion [32]. Hong et al. [33] proposed an encoder–decoder framework for multi-source land-cover classification that sequentially reconstructs the original multimodal inputs from fused features. Wu et al. [34] utilized CNNs as the backbone for feature extraction from HSI and LiDAR modalities and employed a Cross-Channel Reconstruction Module (CCR-Net) to enable mutual information exchange between the modalities. Numerous other CNN-based techniques have been proposed in the literature [35] [36]. However, CNN-based models remain limited by their local receptive fields, CNN-based frameworks showed limitations in modality-specific feature extraction, particularly for HSI, as they often treat each pixel and spectral band with equal importance [37], [38]. To address this limitation, several studies have introduced attention-based mechanisms to assist CNNs in extracting and enhancing modality-specific features. For instance, FusATNet [39] employs self-attention to generate modality-specific attention maps that guide CNN-based feature extraction, followed by cross-attention for feature fusion. In another approach, Fang et al. [29] employed CNNs to extract modality-specific features, subsequently refining them using spatial and spectral enhancement modules (SAEM and SEEM) capable of cross-modal interaction, followed by feature fusion through concatenation. These techniques improved classification performance, However, as these attention modules are built upon convolutional operations, they remain constrained by the locality of convolutional receptive fields, limiting their ability to model long-range dependencies. Moreover, most of these attention

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modules operate at a single-level attention [24], which limits their ability to capture the spectral–spatial variability inherent in HSI data.

B. Transformer-Based Multimodal Fusion

Transformers, known for their strong capability to capture long-range contextual dependencies, have been explored for HSI–LiDAR classification. Xue et al. [40] proposed a three-branch transformer-based network to model the spectral features of HSI and the spatial features of both HSI and LiDAR, where interaction between spectral and spatial features is achieved through a cross-attention fusion mechanism. Similarly, SpectralFormer [23], originally designed for HSI-based land-cover classification, has also been adapted for HSI–LiDAR classification and demonstrates superior performance in global feature learning. However, transformer-based models often struggle with capturing local spatial details, which are essential for understanding fine-grained semantics in land-cover classification. In another work, Feng et al. [41] adopt a patch-based input to the transformer instead of pixel-level representations to improve its ability to capture local spatial information. Nevertheless, such patch-based strategies remain limited in their capacity to jointly model local details and global contextual dependencies. To better integrate local and global features for improved classification performance, several hybrid CNN–transformer architectures have been proposed.

C. Hybrid CNN–Transformer Architectures

In the hybrid feature learning paradigm, several modules have been integrated with transformer architectures, including Fourier [42], morphological [43] and CNN-based modules [44], to enhance feature representation. Collectively, these techniques aim to enrich transformer architectures by incorporating complementary local or structure-aware representations through Fourier, morphological, and CNN-based modules, thereby strengthening overall feature learning. Building on this motivation, hybrid CNN–transformer frameworks have emerged as a promising direction, enabling simultaneous extraction of fine-grained spatial details and long-range dependencies for multimodal feature fusion. Some prominent CNN–transformer–based approaches include ExVit [45], which extends the Vision Transformer (ViT) by integrating separable convolution modules within position-shared transformer branches, and feature fusion across modalities is performed through cross-modal attention. Wang et al. [21] proposed a dual-branch transformer framework, where a group embedding module aids transformer's self-attention for modality-specific feature extraction, and a cross-modal attention block enables exchange of information between modalities. Several other studies have been conducted along similar lines [24], [46], focusing on capturing spatial–spectral features for effective land-cover classification. In another study, Yang et al. [47] used LiDAR tokens as queries to identify keys from HSI, thereby selecting the most prominent HSI bands and reducing feature redundancy caused by the redundant spectral bands in HSI.

However, despite recent advances in CNN–transformer fusion architectures, several important challenges remain in HSI–LiDAR classification. First, many existing methods do not explicitly model the interaction between local spatial patterns and global contextual dependencies, which limits their ability to capture the full spectral–spatial diversity of multimodal remote sensing data. Second, most fusion strategies rely on standard cross-attention mechanisms that lack explicit control over cross-modal information exchange, which can lead to redundant or unstable feature integration when modalities exhibit strong heterogeneity. Addressing these limitations requires a framework that jointly enhances local–global feature representation while regulating multimodal information exchange.

To address these challenges, this work focuses on improving local–global feature representation through a convolution-embedded transformer encoder and enhancing multimodal feature fusion via a residual-gated cross-attention mechanism.

III. Methodology

In this work, we propose LGCF-Net, a unified architecture for HSI and LiDAR classification that enables simultaneous local–global feature extraction and controlled multimodal fusion. Fig. 1 illustrates the overall architecture of the proposed network, which consists of four main components: (1) Spectral–Spatial Transition Refinement (SSTR), (2) LiDAR Feature Enhancement Module (LFEM), (3) a Cross-Modal Convolutional Transformer (CMCT) composed of a Local–Global Transformer Encoder (LGTE) and Residual Gated Cross-Attention (RGCA), and (4) a Classification Token Mixer (CTM). The proposed attention-based SSTR module extracts and refines the HSI feature representation, while LFEM extracts and enhances LiDAR features. Within the CMCT, LGTE learns local–global contextual information, whereas RGCA fuses HSI and LiDAR representations by suppressing redundant features. A lightweight token mixer further refines the HSI and LiDAR classification embeddings. Finally, two modality-specific classifier heads generate logits that are fused by element-wise summation to perform the final classification.

A. Data Preprocessing

The original HSI data can be denoted as $X_H \in \mathbb{R}^{H\times W\times D}$, and the LiDAR data as $X_L \in \mathbb{R}^{H\times W}$, where H and W represent the spatial dimensions (height and width), and D denotes the spectral dimension of the HSI data. HSI typically contains hundreds of spectral bands, many of which are highly redundant and can reduce the discriminative ability of deep learning models while also increasing computational complexity. To address this issue, principal component analysis (PCA) [48] is applied as a dimensionality reduction technique, projecting the original HSI data into a lower-dimensional feature space, thereby reducing redundancy and preserving essential spectral information.After dimensionality reduction, the HSI data is represented as $X_H \in \mathbb{R}^{H\times W\times D_{\mathrm{red}}}$, where a common dimensionality of $D_{\mathrm{red}} = 30$ is retained across all datasets to maintain a consistent preprocessing configuration. As illustrated in Fig. 2, the first 30 principal components

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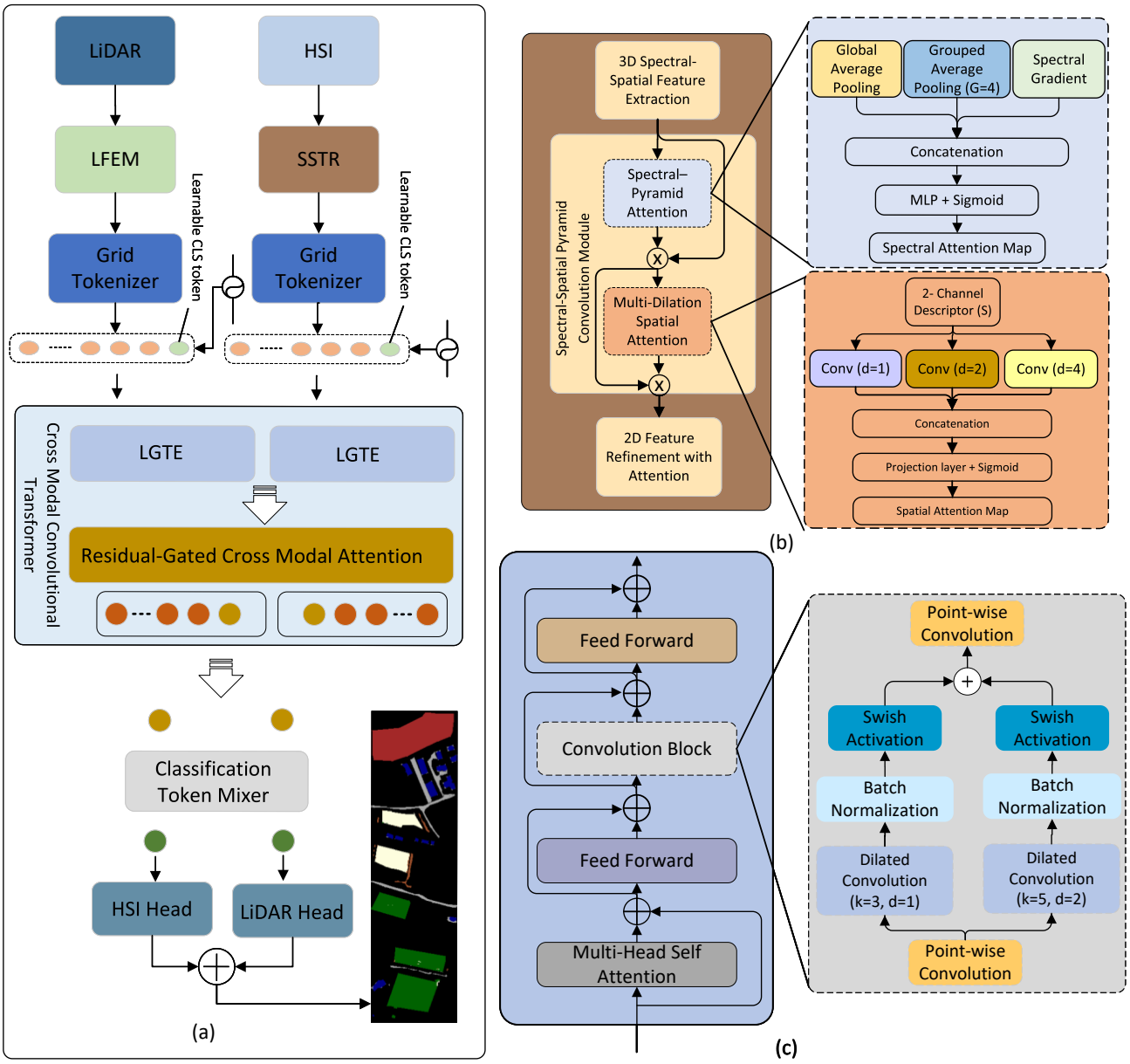


Fig. 1. Architecture of the proposed LGCF-Net: (a) overall workflow of the framework; (b) internal architecture of the Spectral–Spatial Transition Refinement (SSTR), composed of the Spectral-Pyramid Attention (SPyA) and Multi-Dilation Spatial Attention (MdSA) modules, sandwiched between 3D and 2D convolutional refinement blocks; and (c) internal architecture of the Local–Global Transformer Encoder (LGTE), embedded with a multiscale convolutional block to facilitate simultaneous local–global feature interaction.

preserve more than 99.8% of the cumulative variance across all four datasets. The explained-variance curves substantially saturate before 30 components, indicating that this setting retains nearly all spectral information while considerably reducing the original spectral dimensionality.

To preserve boundary information, padding is applied to both the HSI and LiDAR data, and $S \times S$ (where $S = 11$) patches are then generated as input to the model. An 11×11 patch size was selected to balance neighbourhood context and class-specific local information. Smaller patches provide a more limited spatial neighbourhood around the centre pixel, whereas substantially larger patches may include more pixels from adjacent land-cover classes. The central pixel of each patch corresponds to the class label. Consequently, the processed data is represented as $X_H^P \in \mathbb{R}^{S\times S\times D_{\text{red}}}$ for HSI and $X_L^P \in \mathbb{R}^{S\times S\times 1}$ for LiDAR, respectively.

B. Spectral–Spatial Transition Refinement (SSTR)

The SSTR is designed to progressively extract and refine the spectral and spatial features of HSI. The initial feature extraction is performed through 3D Spectral–Spatial Feature Extraction module, followed by a Spectral-Spatial Pyramid Convolutional (SSPC) module. Subsequently, a 2D CNN along with spatial attention is used for further feature enhancement.

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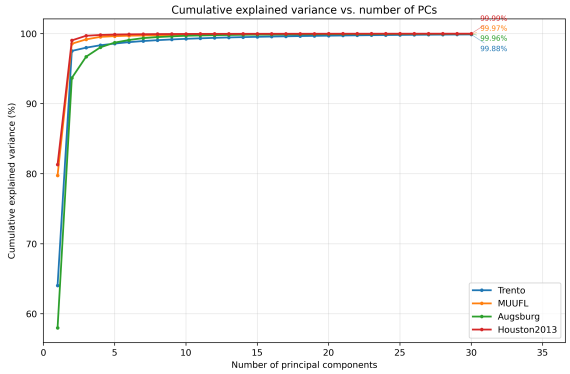


Fig. 2. Cumulative explained variance across principal components, demonstrating the suitability of the selected PCA dimensionality.

1) 3D Spectral–Spatial Feature Extraction: The initial spectral–spatial feature extraction of HSI is performed using a 3D convolutional module consisting of a 3D convolutional layer, normalization, ReLU activation, and dropout. This module serves as the foundation of the HSI processing pipeline by generating an initial spectral–spatial representation. Despite its simplicity, it effectively converts raw HSI data into a compact and informative feature representation, facilitating deeper and more discriminative feature learning in the later stages of the network. The 3D Spectral–Spatial Feature Extraction is formulated as follows:

$$\mathrm{F}_H^{3D} = \mathrm{Dropout}\big(\mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv3D}\big(\mathrm{X}_H^P\big)\big)\big)\big) \tag{1}$$

2) Spectral–Spatial Pyramid Convolution Attention: The features extracted from the previous module are further processed through the Spectral–Spatial Pyramid Convolution (SSPC) module. This module consists of two key components: a Spectral–Pyramid Attention (SPyA) and a Multi-dilation Spatial Attention (MdSA). The detailed structures and functionalities of these components are described in the following subsections.

Spectral–Pyramid Attention (SPyA) To enhance spectral discrimination in HSI feature maps, the proposed SPyA module refines 3-D output F_H^{3D} by jointly modeling global, grouped, and gradient-based spectral statistics. First, global average pooling is applied over the spectral and spatial dimensions to obtain a global descriptor $\mathbf{g} \in \mathbb{R}^{B\times C}$, representing overall channel activation. Independent of the global descriptor, the spectral axis is partitioned into $G = 4$ groups, and average pooling is applied within each group to produce a complementary multi-scale spectral context descriptor $\mathbf{g}_p$, forming a hierarchical "spectral pyramid". In parallel, a depth-wise $3 \times 1 \times 1$ convolution is performed along the spectral dimension to capture local spectral variations, producing a spectral-gradient descriptor $\mathbf{g}_g$. The three descriptors are concatenated and passed through a lightweight two-layer MLP with a sigmoid activation to generate a channel-wise attention map. Mathematically, it is expressed as follows:

$$\tilde{F}_H^{3D} = M_c^{3D} \odot F_H^{3D}, \qquad M_c^{3D} = \sigma(\mathrm{MLP}\left([\mathbf{g} \,\|\, \mathbf{g}_p \,\|\, \mathbf{g}_g]\right)) \tag{2}$$

,

Multi-Dilation Spatial Attention (MdSA) Following the spectral-channel refinement, the MdSA module is designed to model spatial saliency at multiple receptive fields. Given the spectrally enhanced feature tensor $\tilde{F}_H^{3D} \in \mathbb{R}^{B\times C\times D\times H\times W}$, two aggregated spatial descriptors are first computed by channel-wise average and maximum pooling, forming a two-channel spatial descriptor mathematically shown as below:

$$\mathrm{S} = [\mathrm{AvgPool}_c(\tilde{F}_H^{3D});\, \mathrm{MaxPool}_c(\tilde{F}_H^{3D})]. \tag{3}$$

To capture contextual information at different 3-D neighbourhood scales, S is processed by set of 3D convolutional kernels with multiple dilation rates $(1, 2, 4)$. These dilated convolutions enlarge the receptive field without increasing the number of parameters, allowing the model to perceive both local and global spatial dependencies. The resulting feature maps are concatenated and projected through a $1 \times 1 \times 1$ convolution followed by a sigmoid activation to obtain the spatial attention map as

$$\begin{aligned}\mathrm{M}_s^{3D} = \sigma\big(\mathrm{Conv}_{1\times1\times1}([\mathrm{Conv}_{3\times3\times3}^{d=1}(\mathrm{S});\\ \mathrm{Conv}_{3\times3\times3}^{d=2}(\mathrm{S});\mathrm{Conv}_{3\times3\times3}^{d=4}(\mathrm{S})])\big),\end{aligned} \tag{4}$$

The final spatially enhanced feature is then computed as

$$\mathrm{Y}_{\mathrm{H}}^{3\mathrm{D}} = \mathrm{M}_s^{3D} \odot \tilde{F}_H^{3D}, \tag{5}$$

which emphasizes spatially informative regions and suppresses less discriminative areas.

Within the SSTR, the SSPC module serves as a dedicated attention sub-block. It integrates SPyA and MdSA to enhance the feature representation of HSI. The complete SSPC module sequentially applies SPyA and MdSA, yielding the final output $\mathrm{Y}_{\mathrm{H}}^{3\mathrm{D}}$ shown in following equation:

$$\mathrm{Y}_{\mathrm{H}}^{3\mathrm{D}} = \mathrm{M}_s^{3D} \odot (\mathrm{M}_c^{3D} \odot \mathrm{F}_H^{3D}). \tag{6}$$

The output of the SSPC module is flattened $\hat{\mathrm{Y}}_H^{2\mathrm{D}}$ and passed through a 2D convolutional along with Convolutional Block Attention Module (CBAM) style attention [49]. Therefore, SSTR progressively refines the features from 3D to 2D representations, emphasizing informative spectral channels and suppressing redundant information, thereby improving spectral–spatial representation quality prior to tokenization.

$$Z_H^{2D} = \mathrm{Conv2D}\left(\hat{Y}_H^{2D}\right) \tag{7}$$

$$F_H^{2D} = \mathrm{Dropout}\big(\phi\big(\mathrm{GN}\big(Z_H^{2D}\big)\big)\big) \tag{8}$$

The PCA-reduced HSI patch $X_H^P \in \mathbb{R}^{B\times1\times30\times11\times11}$ is first transformed by 3D convolution into $F_H^{3D} \in \mathbb{R}^{B\times8\times28\times9\times9}$. SPyA and MdSA refine this representation without changing its dimensions. The channel and spectral dimensions are then merged to obtain $\hat{Y}_H^{2D} \in \mathbb{R}^{B\times224\times9\times9}$, followed by 2D

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convolution to produce $F_H^{2D} \in \mathbb{R}^{B\times 64\times 9\times 9}$ for subsequent tokenisation and local–global modelling.

Through this hierarchical mechanism, the SSTR jointly enhances spectral and spatial discriminability, producing more informative HSI feature representations.

C. LiDAR Feature Enhancement Module (LFEM)

After data preprocessing, LiDAR features are processed through a 2D CNN module comprising a single 2D convolutional layer, followed by a normalization layer, ReLU activation, and a dropout layer. The features extracted from this module are subsequently passed to a spatial attention module, which further enhances the spatial representations. The spatial attention module is composed of average- and max-pooling operations followed by a convolutional layer, with a sigmoid function used for activation. The architecture of the LFEM is illustrated in Fig. 3 and is mathematically expressed as follows:

$$\mathrm{F}_L^{2D} = \mathrm{Dropout}\big(\mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv2D}\big(\mathrm{X}_L^{P}\big)\big)\big)\big). \quad (9)$$

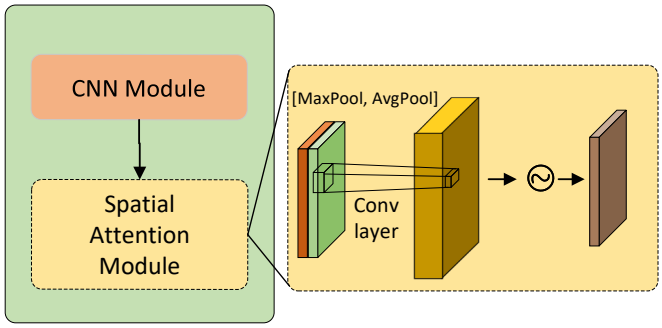


Fig. 3. Architecture of the LiDAR Feature Enhancement Module (LFEM) with CNN-based feature extraction and spatial attention for feature refinement.

D. Grid Tokenization

The 2D feature map is first projected to the transformer embedding dimension using a 1×1 convolution. It is then divided into a fixed $G \times G$ grid (with $G = 3$) via adaptive average pooling, and each grid cell is flattened to form a spatial token. In parallel, a learnable `[CLS]` token is augmented by adding a global context vector obtained through global average pooling of the projected feature map. The `[CLS]` token and grid tokens are concatenated and combined with learnable positional embeddings to construct a compact token sequence for transformer processing. This grid tokenization [50], [51] is applied to both the HSI and LiDAR feature maps, producing token sequences T_H and T_L, respectively:

$$\mathrm{T}_H = [\mathrm{Cls}_H;\, \mathrm{T}_H] + \mathrm{P}_H, \quad (10)$$

$$\mathrm{T}_L = [\mathrm{Cls}_L;\, \mathrm{T}_L] + \mathrm{P}_L, \quad (11)$$

where Cls_H, Cls_L denotes classification tokens and P_H, P_L denotes corresponding positional embeddings for HSI and LiDAR modalities.

E. Cross-Modal Convolutional Transformer (CMCT)

The proposed CMCT consists of two stages. In the first stage, local–global features are extracted from both HSI and LiDAR modalities using LGTE, i.e., a multiscale convolutional embedded transformer encoder. In the second stage, the fusion of these modalities is performed through a residual gated cross-attention mechanism (RGCA).

1) Local-Global Transformer Encoder: The tokens produced by the grid tokenizer are subsequently utilized to model semantic relationships among feature tokens through the multiscale convolutional embedded transformer. In a transformer, the multi-head self-attention mechanism serves as the core component for capturing global semantic dependencies between tokens. It consists of multiple self-attention heads, each of which computes correlations among tokens using three learnable projection matrices. These matrices transform the feature tokens into three distinct representations: Q (query), K (key), and V (value). The attention scores are then derived from Q and K through a softmax operation [50]. The self-attention operation is mathematically defined as:

$$\mathrm{SA} = \mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{D}}\right) V. \quad (12)$$

In multi-head self-attention, the feature dimension D is divided across h attention heads such that $D = \frac{C}{h}$, where C denotes the embedding dimension and h represents the number of heads. Each head independently performs attention computation on a lower-dimensional subspace, allowing the model to capture diverse feature relationships. The outputs from all heads are then concatenated and projected back to the original dimensionality through a linear transformation using a projection matrix W^O. This operation restores the complete representation while integrating complementary information from different heads. The multi-head self-attention is formulated as

$$\mathrm{MHSA}(Q, K, V) = \mathrm{Concat}(\mathrm{SA}_1, \mathrm{SA}_2, \ldots, \mathrm{SA}_h) W^O, \quad (13)$$

where $\mathrm{SA}_i = \mathrm{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i$ represents the attention output of the i-th head, and Q_i, K_i, and V_i denote the respective query, key, and value projections for that head. Although MHSA effectively captures global dependencies among tokens, it does not explicitly model local semantic correlations, which limits the ability of transformers to capture local details.

Unlike conventional transformers that primarily capture global dependencies among feature tokens [44], [52], [53], we incorporate dilated convolutions within the transformer encoder to jointly model both local and global relationships sequentially. The convolutional block is positioned between two feed-forward layers, and a residual connection is introduced to prevent vanishing gradients during training. The convolutional block comprises a point-wise convolution followed by two

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parallel 1D dilated convolutional layers with kernel sizes of three and five, respectively, to capture multiscale contextual information. Each convolutional branch is followed by batch normalization and a Swish activation function to enhance nonlinearity and stabilize feature learning. The outputs from the two parallel branches are summed to enrich feature representation, and a final point-wise convolution is applied to project the features back to their original dimensionality. It is represented mathematically as follows:

$$\mathrm{H}_1 = \mathrm{H}^{(l)} + \mathrm{MHSA}\Big(\mathrm{LN}(\mathrm{H}^{(l)})\Big), \tag{14}$$

$$\mathrm{U} = \mathrm{H}_1 + \mathrm{FFN}(\mathrm{LN}(\mathrm{H}_1)), \tag{15}$$

$$\mathrm{H}_2 = \mathrm{H}_1 + \mathcal{C}(\mathrm{U}), \tag{16}$$

$$\mathcal{C}(\mathrm{U}) = W_o * \phi\Big(\mathrm{Conv}_{k=3}^{d_1}(\mathrm{U}) + \mathrm{Conv}_{k=5}^{d_2}(\mathrm{U})\Big), \tag{17}$$

$$\mathrm{H}^{(l+1)} = \mathrm{H}_2 + \mathrm{FFN}(\mathrm{LN}(\mathrm{H}_2)), \tag{18}$$

where $\mathrm{H}^{(l)}$ input tokens to layer l, H_1 and H_2 represents outputs after the MHSA and the convolution block $\mathcal{C}(U)$, respectively.

2) Residual Gated Cross-Attention Mechanism: Both HSI and LiDAR modalities are first processed independently to obtain modality-specific representations. While the two modalities offer complementary information, conventional cross-attention mechanisms often pass redundant or weakly relevant features between modalities, which can degrade classification performance. To address this, we introduce the Residual-Gated Cross-Attention (RGCA) mechanism, which forms the central component of our fusion design.

RGCA incorporates a learnable gating parameter that regulates the flow of cross-modal information through a sigmoid gate, enabling the model to emphasize informative complementary features while suppressing redundant features. In RGCA, the CLS token of one modality attends to all tokens of the other modality to exchange complementary information. The gated residual connection ensures that only discriminative and relevant features are incorporated. The updated CLS token is then projected back to its own feature tokens, promoting intra-modal consistency. Because each CLS token has already been enriched by the LGTE, this step further strengthens the discriminative capacity of the fused representation. Fig. 4 provides an overview of the RGCA mechanism, where each modality contains feature tokens and a CLS token. During fusion, the CLS token from one modality replaces the CLS token of the other modality, forming a set of cross-modal tokens that is subsequently processed by the cross-attention module. The cross-attention operation is parameterized by learnable matrices W_q, W_k, and W_v. Similar to self-attention, multi-head cross-attention is employed; cross-attention is performed between $x_{\mathrm{Cls}}^{hsi'}$ and X_{ct}^{hsi}. The CLS token $x_{\mathrm{Cls}}^{hsi'}$ is the query used for cross-attention that contains information of the HSI feature tokens. The cross-attention is mathematically

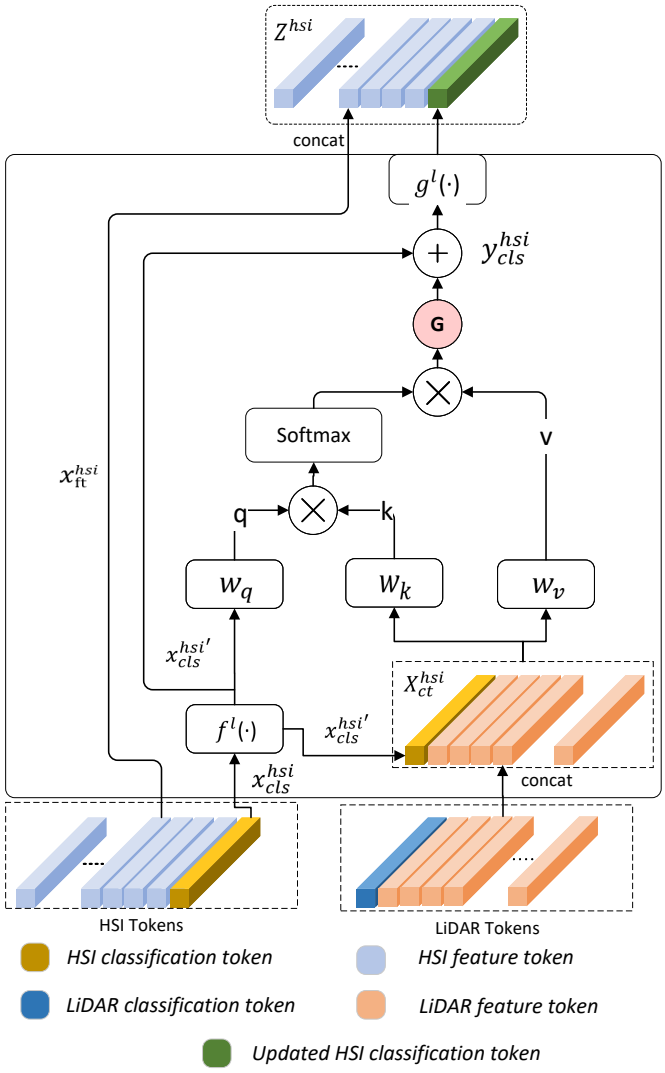


Fig. 4. Residual-gated cross-modal attention. The HSI CLS token attends to concatenated cross-modal tokens to compute an attention update, which is modulated by a gating function G and added via a residual connection to produce the updated CLS representation.

represented in equation (19), while the final output of the RGCA module is shown in equations 20 and 21.

$$\begin{aligned} &q = x_{\mathrm{cls}}^{hsi'} W_q, \quad k = X_{\mathrm{ct}}^{hsi} W_k, \quad v = X_{\mathrm{ct}}^{hsi} W_v, \\ &\mathbf{A} = \mathrm{softmax}\left(\frac{qk^T}{\sqrt{C/h}}\right), \quad \mathrm{CA}(X_{\mathrm{ct}}^{hsi}) = \mathbf{A}v, \end{aligned} \tag{19}$$

$$y_{\mathrm{cls}}^{hsi} = f^l(x_{\mathrm{cls}}^{hsi}) \; + \; \sigma(\gamma_{\mathrm{h}})\,\mathrm{MHCA}(\mathrm{LN} f^l(X_{\mathrm{ct}}^{\mathrm{hsi}})), \tag{20}$$

The gating parameter is a learnable scalar, not a predefined or class-specific threshold. It controls the overall strength of cross-modal information exchange at each RGCA stage rather than assigning modality dependence to individual land-cover classes.

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$$Z^{hsi} = [g^l(y^{hsi}_{\text{cls}}) \parallel x^{hsi}_{\text{ft}}], \tag{21}$$

where $f^l(\cdot)$ is the linear transformation on the HSI CLS token for dimensionality alignment and $\sigma(\gamma_{\text{h}})$ is the sigmoid applied to the learnable scalar gating parameter. Similarly, the output of cross attention for LiDAR is represented as follows:

$$Z^{lidar} = [g^l(y^{lidar}_{\text{cls}}) \parallel x^{lidar}_{\text{ft}}]. \tag{22}$$

F. Classification Token Mixer

The post-fusion mixer refines the classification tokens obtained through cross-modal fusion. The architecture of the module is shown in Fig. 5. It is a lightweight, convolution-based module that encourages bi-directional exchanges between modalities, thereby strengthening the complementary information present in them. The refined CLS representations are subsequently processed through modality-specific classification heads, where independent normalization and linear layers produce modality-wise logits. It is mathematically represented as follow:

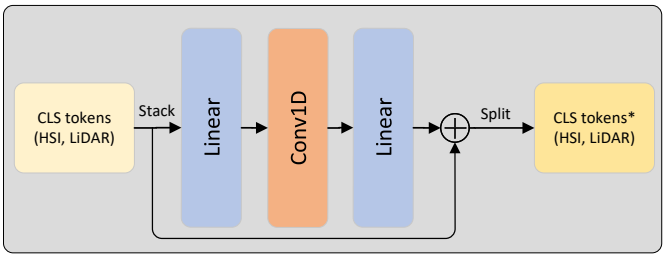


Fig. 5. The architecture of the classification token mixer.

$$\tilde{\text{z}}_h = \text{LN}(\text{Z}^{hsi}), \qquad \tilde{\text{z}}_l = \text{LN}(\text{Z}^{lidar}), \tag{23}$$

where $\text{LN}(\cdot)$ denotes layer normalization. Each normalized token is passed through a fully connected layer to produce class logits:

$$\hat{\text{y}}_h = \tilde{\text{z}}_h \text{W}_h + \text{b}_h, \qquad \hat{\text{y}}_l = \tilde{\text{z}}_l \text{W}_l + \text{b}_l, \tag{24}$$

where $\text{W}_h, \text{W}_l \in \mathbb{R}^{D\times C}$ are the learnable weight matrices, $\text{b}_h, \text{b}_l \in \mathbb{R}^C$ are bias vectors, and C is the number of classes.

The final classification is performed through the summation of modality-specific logits, which preserves modality-specific characteristics and thus improves the overall classification performance:

$$\hat{\text{y}} = \hat{\text{y}}_h + \hat{\text{y}}_l. \tag{25}$$

During training, the class probabilities are computed by applying the softmax:

$$p_i = \frac{e^{\hat{y}_i}}{\sum_{j=1}^{C} e^{\hat{y}_j}}, \tag{26}$$

where p_i denotes the predicted probability for class i.

Algorithm 1 Proposed LGCF-Net Training and Inference Procedure

Input: HSI data $\text{X}_{HSI} \in \mathbb{R}^{H\times W\times C}$, LiDAR data $\text{X}_{LiDAR} \in \mathbb{R}^{H\times W}$, ground truth $GT \in \mathbb{R}^{H\times W}$, PCA bands $b = 30$, patch size $P \times P$, training ratio ψ
Output: Predicted labels of the test dataset
Set batch size $= 64$, optimizer = Adam (learning rate $= 10^{-3}$), epochs $= 100$
Obtain X_{HSI} after PCA transformation
Create patch samples for HSI and LiDAR data and split them into training and testing sets
Generate training and testing data loaders
for $i = 1$ to e **do**
    Extract features using the SSTR module for HSI
    Extract features using the LFEM module for LiDAR
    Perform grid tokenisation and initialize weights to generate tokens
    Concatenate learnable tokens and add positional embeddings
    Apply transformer encoder (LGTE) for local-global feature extraction
    Apply residual gated cross attention (RGCA) for cross modal alignment
    Process aligned features through classification token mixer
    CLS tokens passed through separate classification heads before summation and softmax to predict class labels
end for
Evaluate the test dataset using the trained model

IV. Experimental Results

This section provides a brief description of the datasets used to evaluate the proposed technique, followed by the experimental setup. Subsequently, comparisons with state-of-the-art methods across different datasets are presented and discussed.

A. Dataset Description

1) MUUFL Dataset: The MUUFL dataset was collected over the University of Southern Mississippi Gulf Park campus in November 2010. The dataset includes two modalities, HSI And LiDAR, with spatial size of 325 $\times$ 220. The HSI data consists of 72 spectral bands, while the LiDAR data has two rasters providing elevation information [54].

2) Trento Dataset: This is one of the most widely used datasets collected over the southern region of Trento, Italy. It comprises six classes and two modalities. The HSI data were acquired using the AisaEAGLE sensor and consist of 63 spectral bands with wavelengths ranging from 0.43 μm to 0.99 μm. The data have a spatial resolution of 1 m, covering an area of 600×166 pixels. The LiDAR data were collected using the Optech ALTM 3100EA sensor [55].

3) Augsburg Dataset: The Augsburg dataset was collected over the city of Augsburg, Germany, using HSI and LiDAR modalities. The dataset covers an area of 332×485 pixels. The

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TABLE I
CHARACTERISTICS OF THE MUUFL GULFPORT, TRENTO, AUGSBURG, AND HOUSTON HYPERSPECTRAL–LIDAR DATASETS.

Dataset		MUUFL	Trento	Augsburg	Houston
Classes		11	6	7	15
Spatial size		325×220	600×166	332×485	349×1905
Spectral bands	HSI	72	63	180	144
	LiDAR	2	1	1	1
Spectral range (μm)		0.38–1.05	0.42–0.99	0.40–2.50	0.38–1.05
Labelled samples		53,687	30,214	78,294	15,029

HSI and LiDAR data were acquired using the HySpex sensor and the DLR-3k sensor, respectively. The HSI data consist of 180 spectral bands with wavelengths ranging from 0.4 μm to 2.5 μm [56].

4) Houston Dataset: The Houston dataset was collected over the University of Houston campus and its surrounding urban area. It consists of a hyperspectral image and a co-registered LiDAR-derived digital surface model (DSM), both with a spatial resolution of 2.5 m. The hyperspectral image contains 144 spectral bands covering the wavelength range of 0.38 μm to 1.05 μm and is calibrated to at-sensor spectral radiance units, while the DSM provides elevation information above sea level. The spatial size of the data is 349×1905 pixels, and the dataset contains 15 land-cover classes [57], [58].

B. Experimental Setup

The proposed model was implemented using the PyTorch framework. The Adam optimizer was used with an initial learning rate of 10^{-3}. The model was trained for 100 epochs with a batch size of 64 using the cross-entropy loss function. Each experiment was repeated five times, and the average results are reported.

Following the experimental convention adopted in previous HSI–LiDAR classification studies [22], [59], [7] no separate validation set was used. Labelled centre pixels were assigned to separate training and test sets, ensuring that the same labelled pixel was not used in both sets. As commonly adopted in patch-based HSI–LiDAR classification, patches centred on nearby pixels may partially overlap. The proposed method was compared with six state-of-the-art transformer-based hybrid methods. All methods were re- evaluated using the same data split, training samples, patch-generation procedure, and experimental settings to ensure a fair comparison. Performance was assessed using Overall Accuracy (OA), Average Accuracy (AA), and the Kappa coefficient.

C. Results and Discussion

This section presents the comparative performance analysis of the proposed method against six state-of-the-art transformer-based hybrid methods. The analysis considers both quantitative evaluation metrics and visual results, to provide a comprehensive assessment of classification performance and feature discriminability.

1) Results on MUUFL Dataset: Table II reports the per-class classification accuracy on the MUUFL dataset. The proposed LGCF-Net achieves the highest overall accuracy of 91.19%, outperforming the strongest baseline (MCFNet) by 1.84%. In addition, LGCF-Net achieves the best performance on several challenging classes such as Trees, Mixed Ground Surface, and Road, indicating the effectiveness of the proposed local–global feature modelling and residual-gated cross-modal fusion mechanisms. These results demonstrate that the proposed architecture effectively integrates spectral and structural information from HSI and LiDAR modalities. LGCF-Net shows comparatively lower performance on the Sidewalk class. Its narrow spatial extent means that the surrounding 11×11 patches may contain substantial information from neighbouring classes such as Road and Building. Consequently, even with local–global modelling, contextual aggregation may dilute the discriminative information of this narrow structure. Although Yellow Curb achieves 96.96% accuracy, this class contains only 33 test samples, making its class-wise accuracy sensitive to a small number of misclassifications. A comparison of classification maps on the MUUFL dataset is presented in Fig. 6.

2) Results on Trento Dataset: Table III reports the classification results on the Trento dataset. The Trento dataset is relatively less challenging than MUUFL due to its clearer spectral separability, simpler class structure and high spatial resolution, resulting in near-saturated performance on certain classes (e.g., *Ground* and *Wood*). However, the proposed technique attains consistent performance across all the classes including challenging classes such as Apple Trees and Buildings. This balanced class-wise behavior is further reflected in overall performance metrics indicating robustness and stable performance. A visual comparison of all the compared techniques on Trento dataset is illustrated in Fig. 7.

3) Results on Augsburg Dataset: Table IV reports the classification results on the Augsburg dataset. Class-wise analysis shows that the proposed approach attains the highest accuracy on the Forest and Industrial Area classes and remains competitive across all other categories. A clear trend can be observed for the Industrial Area and Commercial Area classes, which are particularly challenging due to their similar structural characteristics and closely overlapping spectral responses. All techniques either perform well on one of these classes while exhibiting degraded performance on the other. In contrast, the proposed method handles these challenging classes more effectively and demonstrates consistent performance across both classes, achieving a notable improvement of approximately 5.93% on the Industrial Area class. HCT achieves the highest accuracy (91.12%) on the Commercial Area class but attains only 46.56% accuracy on the Industrial Area class, clearly demonstrating its limitation in distinguishing between these similar classes. In contrast, the proposed method achieves 72.71% and 73.77% accuracy on the Industrial Area and Commercial Area classes, respectively. The balanced and consistent performance across all classes results in the proposed method achieving the highest overall accuracy (93.96%) and Kappa coefficient (91.40%), demonstrating superior performance compared with existing

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TABLE II
PER-CLASS CLASSIFICATION ACCURACY (%) ON THE MUUFL GULFPORT DATASET. THE BEST RESULTS ARE SHOWN IN BOLD.

No.	Class (Train/Test)	HCT [25]	ExViT [45]	MFT [22]	CrossHL [7]	CET [60]	MCFNet [59]	LGCF-Net (Ours)
1	Trees (150/23096)	86.10	88.30	89.69	89.12	88.31	89.92	**91.73**
2	Mostly Grass (150/4120)	**90.49**	88.88	80.63	88.20	84.78	88.05	86.79
3	Mixed Ground Surface (150/6732)	72.89	81.26	79.05	78.50	73.98	82.99	**85.32**
4	Dirt and Sand (150/1676)	91.28	95.94	93.73	96.28	93.67	97.25	**97.19**
5	Road (150/6537)	85.31	90.16	75.24	88.95	75.41	86.87	**93.94**
6	Water (150/316)	**100**	**100**	99.36	99.81	**100**	**100**	**100**
7	Building Shadow (150/2083)	90.49	**97.55**	91.59	93.89	90.87	93. 95	90.06
8	Building (150/6090)	94.89	95.13	93.66	**96.38**	93.64	95.63	96.35
9	Sidewalk (150/1235)	70.28	70.85	70.44	**81.50**	78.21	77.32	78.21
10	Yellow Curb (150/33)	87.87	90.90	90.90	96.96	**96.96**	96.96	96.96
11	Cloth Panels (150/119)	99.15	95.18	96.63	99.49	**100**	100	100
Metrics	**OA (%)**	85.76	88.29	86.07	88.84	85.05	89.35	**91.19**
	AA (%)	88.08	89.93	87.36	91.74	88.71	91.72	**92.41**
	Kappa (%)	81.59	84.71	81.82	85.44	80.61	86.07	**88.43**

TABLE III
PER-CLASS CLASSIFICATION ACCURACY (%) ON THE TRENTO HYPERSPECTRAL–LIDAR DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

No.	Class (Train/Test)	HCT [25]	ExViT [45]	MFT [22]	CrossHL [7]	CET [60]	MCFNet [59]	LGCF-Net (Ours)
1	Apple Trees (129/3905)	98.92	99.43	98.27	99.47	98.46	**99.97**	99.79
2	Buildings (125/2778)	98.66	99.10	97.71	98.73	97.87	97.62	**99.49**
3	Ground (105/374)	**100**	**100**	99.47	**100**	**100**	**100**	**100**
4	Wood (154/8969)	**100**	**100**	**100**	**100**	**100**	**100**	**100**
5	Vineyard (184/10317)	**100**	**100**	98.98	99.92	99.77	99.99	**100**
6	Roads (122/3052)	97.60	99.01	99.35	99.59	**99.62**	98.68	98.00
Metrics	**OA (%)**	99.48	99.62	99.37	99.59	99.38	99.61	**99.71**
	AA (%)	99.20	99.36	98.36	99.39	99.29	99.37	**99.54**
	Kappa (%)	99.30	99.55	99.15	99.46	99.17	99.48	**99.62**

TABLE IV
PER-CLASS CLASSIFICATION ACCURACY (%) ON THE AUGSBURG HYPERSPECTRAL–LIDAR DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

No.	Class (Train/Test)	HCT [25]	ExViT [45]	MFT [22]	CrossHL [7]	CET [60]	MCFNet [59]	LGCF-Net (Ours)
1	Forest (1162/22084)	99.32	99.32	98.40	99.62	99.48	99.36	**99.71**
2	Residential Area (344/6538)	89.38	87.29	89.48	89.27	93.17	**94.31**	92.57
3	Industrial Area (91/1735)	46.56	56.59	66.78	58.38	40.71	56.22	**72.71**
4	Low Plant (334/6353)	**98.58**	98.03	95.23	96.74	96.02	96.26	97.64
5	Allotment (20/555)	78.26	82.60	79.65	80.47	**96.73**	73.91	78.07
6	Commercial Area (23/443)	**91.12**	41.74	80.61	59.13	90.21	63.27	73.77
7	Water (312/5928)	87.43	86.28	**93.66**	86.19	80.78	84.72	87.84
Metrics	**OA (%)**	92.02	90.58	92.20	91.31	93.40	93.00	**93.96**
	AA (%)	84.38	77.83	**86.26**	81.40	85.30	81.15	86.05
	Kappa (%)	88.70	86.66	89.09	87.70	90.61	90.23	**91.40**

transformer-based hybrid techniques. A visual comparison of all the compared techniques on Augsburg dataset is illustrated in Fig. 8.

4) Results on Houston Dataset: The Houston dataset [57] presents a challenging multimodal benchmark with 15 land-cover classes over a complex urban scene. To assess the robustness of LGCF-Net under limited training samples, we

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TABLE V
CLASS-WISE ACCURACIES (%) AND OVERALL EVALUATION METRICS OF DIFFERENT METHODS ON THE HOUSTON DATASET. BEST RESULTS ARE HIGHLIGHTED IN BOLD.

No.	Class (Train/Test)	HCT [25]	ExViT [45]	MFT [22]	CrossHL [7]	CET [60]	MCFNet [59]	LGCF-Net (Ours)
1	Healthy grass (20/1231)	91.87	93.33	**98.70**	93.14	95.53	97.96	95.20
2	Stressed grass (20/1234)	87.84	93.76	98.05	98.55	98.62	**98.94**	98.13
3	Synthetic grass (20/677)	**100**	**100**	**100**	99.94	**100**	**100**	**100**
4	Trees (20/1224)	99.59	99.75	97.71	97.54	99.50	**100**	98.85
5	Soil (20/1222)	**100**	**100**	99.91	**100**	99.75	**100**	**100**
6	Water (20/305)	98.03	95.40	**100**	98.95	93.11	89.18	98.68
7	Residential (20/1248)	93.02	94.55	83.65	89.19	**97.35**	91.90	87.74
8	Commercial (20/1224)	79.41	84.23	87.50	86.99	90.11	81.94	**91.42**
9	Road (20/1232)	83.44	68.01	68.66	81.44	74.83	**88.55**	81.00
10	Highway (20/1207)	94.53	90.80	87.82	89.74	90.88	95.02	**99.17**
11	Railway (20/1215)	92.59	95.06	96.29	93.81	92.01	**97.28**	89.46
12	Parking lot 1 (20/1213)	80.29	88.37	64.30	87.27	90.27	85.16	**90.43**
13	Parking lot 2 (20/449)	**98.66**	89.53	94.20	94.52	95.10	97.32	97.55
14	Tennis court (20/408)	**100**	99.01	99.75	99.80	**100**	**100**	98.52
15	Running track (20/640)	99.37	98.28	**100**	**100**	97.51	99.84	99.84
Metrics	**OA (%)**	91.79	91.81	90.06	92.95	93.69	**94.43**	94.13
	AA (%)	93.24	92.67	91.77	94.06	94.30	94.87	**95.07**
	Kappa (%)	91.12	91.81	89.25	92.38	93.17	93.27	**93.65**

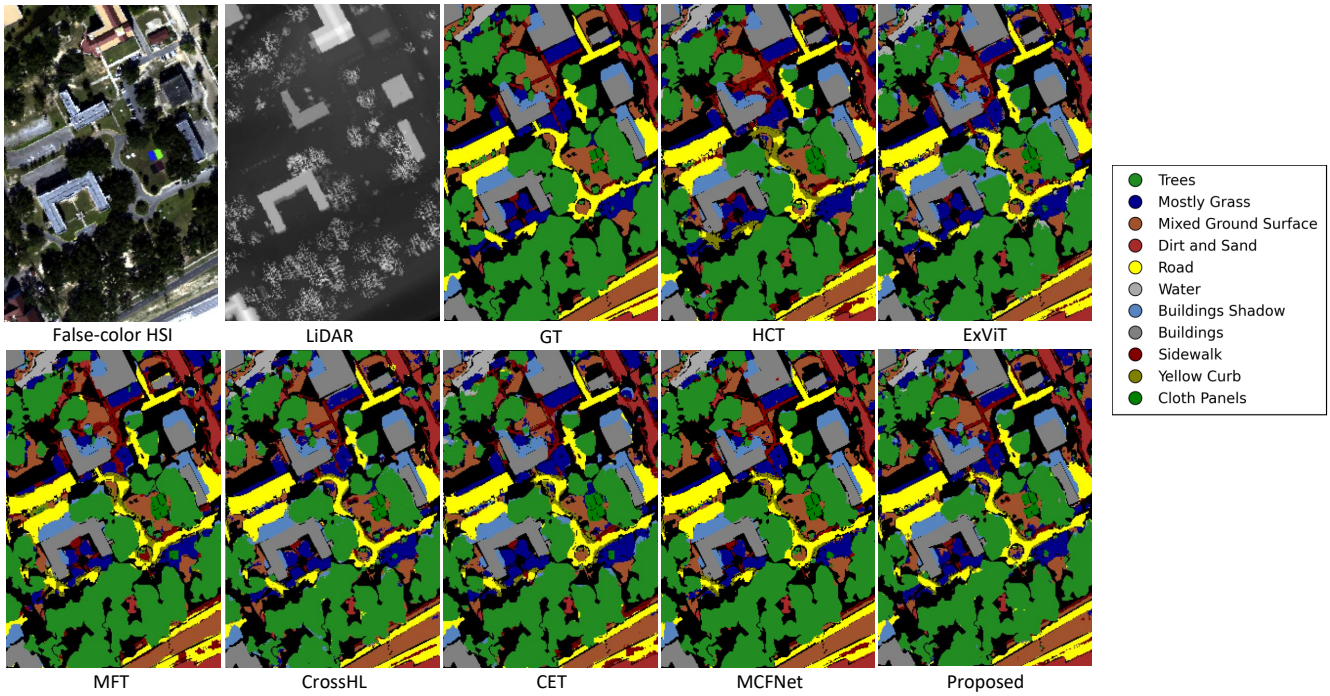


Fig. 6. Qualitative results illustrated through classification maps on the MUUFL test dataset using different methods.

conduct experiments using only 20 training samples per class. All six methods are retrained under identical conditions to ensure a fair comparison, and results are reported in Table V. Among the fifteen classes, residential, commercial, roads, parking lot 1, and parking lot 2 are particularly challenging, as all compared models struggle with these classes. Nevertheless, LGCF-Net technique achieved the highest accuracy on commercial areas with 91.42% and parking lot 1 with 90.43%, while attaining the second best accuracy on parking lot 2 with 97.55%, compared to 98.66% achieved by HCT. Moreover, for the Road and Residential classes, LGCF-Net achieves accuracies of 81.00% and 87.74%, respectively. These findings demonstrate that LGCF-Net method achieves competitive and balanced per-class performance under limited training setting.

5) Overall Result Analysis: On MUUFL and Trento, the proposed technique consistently achieves superior performance across the overall evaluation metrics, while on Augsburg it remains competitive in AA and achieves the highest

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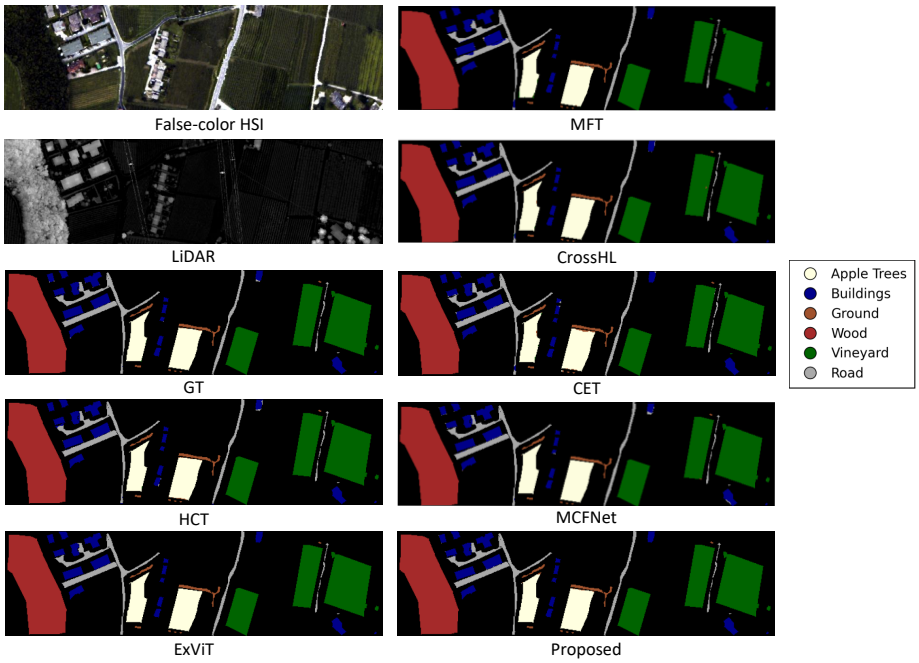


Fig. 7. Qualitative results illustrated through classification maps on the Trento test dataset using different methods.

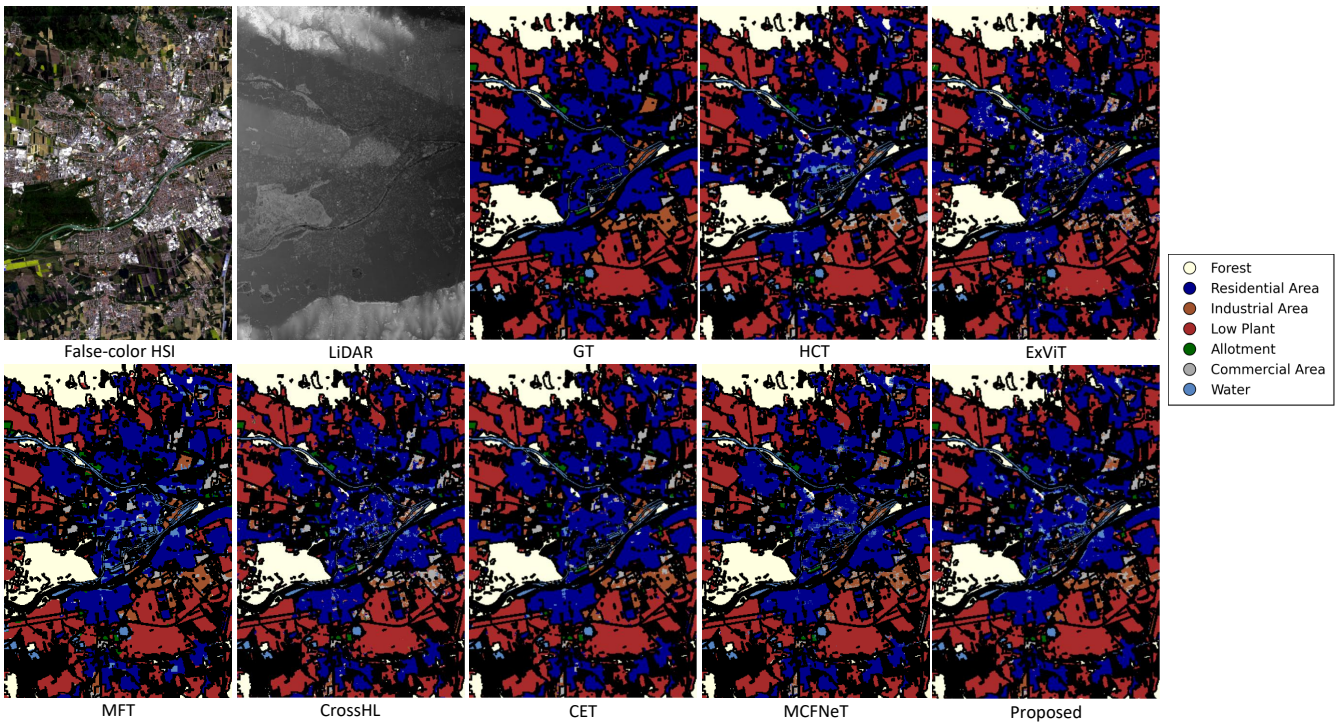


Fig. 8. Qualitative results illustrated through classification maps on the Augsburg test dataset using different methods.

OA and Kappa among the compared transformer-based hybrid methods. On the MUUFL dataset, MCFNet ranks second to the proposed approach, while on the Trento dataset, ExViT demonstrates competitive performance in overall metrics. On the Augsburg dataset, although MFT attains slightly higher (0.21%) average accuracy, the proposed technique achieves better performance in terms of overall accuracy and Kappa coefficient. The discriminative capability of the proposed

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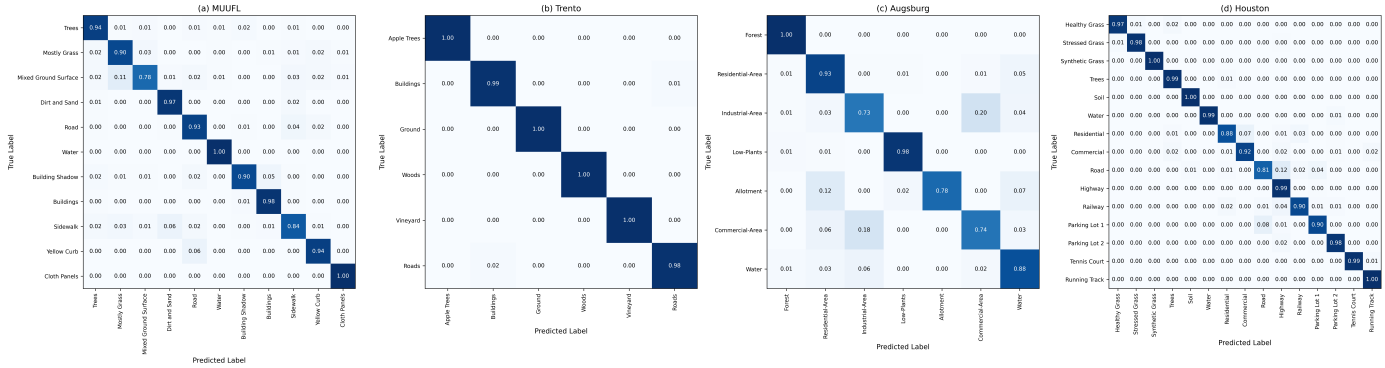


Fig. 9. Normalized confusion matrices of LGCF-Net on the MUUFL, Trento, Augsburg, and Houston datasets for the best-performing run.

technique across the three datasets is also illustrated through t-SNE plots, which also supports quantitative results presented in Table II - IV.

On the Houston dataset, even under the limited training-sample setting, LGCF-Net demonstrates superior performance among the compared methods in terms of AA and the Kappa coefficient. However, in terms of OA, LGCF-Net slightly underperforms MCFNet by 0.3%. This difference arises because OA is influenced by the number of test samples in each class, whereas AA assigns equal importance to all classes. Although LGCF-Net achieves substantial improvements on classes such as Water, Commercial, Highway, and Parking Lot 1, its comparatively lower performance on classes with larger test samples, including Residential, Road, and Railway, has a greater influence on OA. Consequently, LGCF-Net achieves a marginally lower OA while attaining higher AA and Kappa, indicating more balanced class-wise performance.

Fig. 9 presents the normalized confusion matrices of LGCF-Net on the four datasets, providing a detailed view of class-wise prediction patterns and misclassification directions. Most classes exhibit strong diagonal responses, indicating consistent classification performance. The remaining errors are mainly concentrated among classes with similar spectral or structural characteristics. For example, on MUUFL, Mixed Ground Surface exhibits greater confusion with Mostly Grass, whereas on Augsburg, Industrial Area shows noticeable confusion with Commercial Area. A category-level examination further illustrates the complementary nature of HSI and LiDAR information. In the MUUFL dataset, Trees and Mostly Grass represent vegetation classes with potentially similar spectral characteristics but different vertical structures. In the best-performing run shown in Fig. 9, the proposed method achieves per-class classification accuracies of 94% and 90%, respectively, with only limited mutual confusion (1% and 2%). This behaviour is consistent with the benefit of combining LiDAR-derived structural information with HSI spectral features. Similarly, in the best-performing run shown in Fig. 9, Healthy Grass and Stressed Grass exhibit similar low-height structural characteristics but differ spectrally. The proposed method achieves per-class classification accuracies of 97% and 98%, respectively, with negligible mutual confusion. These results are consistent with the fused representation effectively using complementary spectral and structural cues to distinguish challenging land-cover categories.

Fig. 10 presents the OA distribution across five runs for each dataset. The compact box plots and close mean–median values indicate stable performance across repeated experiments.

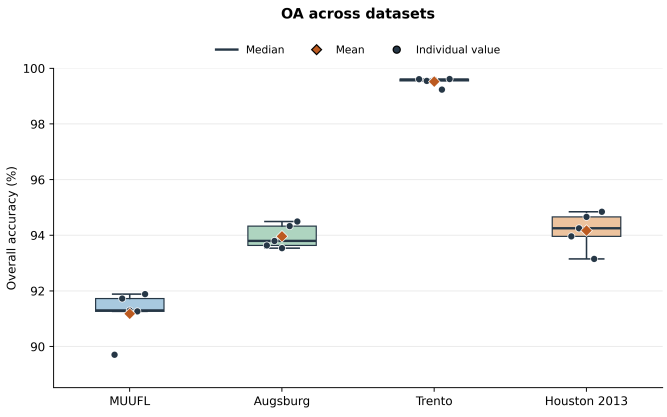


Fig. 10. Distribution of OA values across five independent runs on the MUUFL, Augsburg, Trento, and Houston datasets.

These results indicate that the proposed method maintains stable and generalisable performance across all datasets with varying characteristics and levels of complexity, thereby demonstrating its robustness.

6) Computational Complexity Analysis: To assess the computational efficiency of the proposed framework, Table VI compares the number of parameters, GFLOPs, inference time, training time, and GPU memory usage with representative baseline methods. Although LGCF-Net has the highest parameter count, its computational cost remains moderate at 0.0327 GFLOPs, which is lower than ExViT, MCFNet, and Cross-HL. LGCF-Net requires 26.1 s for inference, which is comparable to MFT and lower than Cross-HL and MCFNet. Inference time denotes full-scene prediction over all image pixels. It also achieves the lowest training time of 280.2 s among the compared methods while maintaining moderate peak GPU memory usage. Overall, these results indicate that the proposed framework maintains manageable computational complexity relative to its classification performance.

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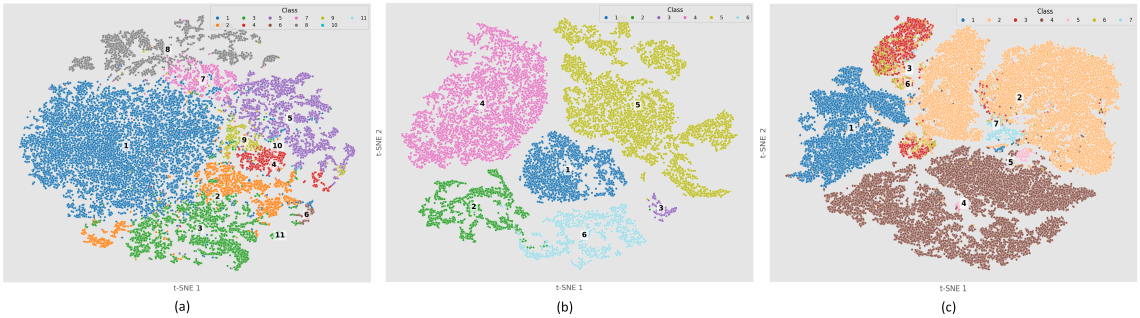


Fig. 11. t-SNE-based feature visualizations on the test data across three datasets: (a) MUUFL, (b) Trento, and (c) Augsburg.

TABLE VI
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT METHODS.

Method	Params (M)	GFLOPs	Infer. (s)	Train (s)	Peak GPU Mem. (MiB)
ExViT	0.234	0.1252	22.6	360.3	40.03
MFT	0.231	0.0118	26.5	370.2	21.90
Cross-HL	0.374	0.0627	53.4	452.5	27.00
MCFNet	0.182	0.0489	46.4	480.5	24.54
Ours	0.654	0.0327	26.1	280.2	30.07

TABLE VII
MODULE-WISE AND FINE-GRAINED ABLATION STUDIES ON MUUFL AND HOUSTON.

Case	Modules			MUUFL			Houston		
	SSTR	RGCA	CTM	OA	AA	κ	OA	AA	κ
(a) Module-wise ablation									
1	×	×	×	85.15	87.98	80.80	91.69	92.81	91.02
2	×	✓	×	87.05	89.57	83.15	92.72	93.97	91.05
3	×	✓	✓	89.51	92.12	86.51	92.62	94.23	91.69
4	✓	✓	✓	**91.19**	**92.41**	**88.43**	**94.13**	**95.07**	**93.65**
(b) Fine-grained SSTR ablation									
SSTR w/o SPyA				89.47	91.40	86.18	92.07	92.90	91.42
SSTR w/o MdSA				88.36	90.90	84.80	92.75	93.10	92.09
Full SSTR				**91.19**	**92.41**	**88.43**	**94.13**	**95.07**	**93.65**

V. ABLATION AND FEATURE REPRESENTATION ANALYSIS

This section further investigates the proposed architecture from three complementary perspectives. First, ablation studies evaluate the contribution of the proposed modules. Second, a parameter analysis examines the model capacity of CMCT under different interaction mechanisms. Third, modality alignment and feature discriminability are analysed using cosine similarity and Linear Discriminant Analysis (LDA), respectively.

A. Module-wise Ablation

A comprehensive ablation study is conducted on the MUUFL and Houston datasets to evaluate the contribution of each core module of LGCF-Net. All configurations are evaluated under identical experimental settings, and the results are reported in Table VII. Four configurations are considered by progressively incorporating RGCA, CTM, and SSTR while preserving the overall network structure.

In Case 1, all three proposed modules are removed to form the baseline configuration. Simple 3D and 2D convolutional layers are used in place of SSTR, conventional cross-attention is used instead of RGCA, and feature summation is used instead of CTM. On MUUFL, the baseline achieves 85.15% OA, 87.98% AA, and 80.80% Kappa, while on Houston it obtains 91.69% OA, 92.81% AA, and 91.02% Kappa.

Case 2 evaluates the contribution of RGCA. Case 1 uses conventional cross-attention without residual gating. Case 2 replaces it with the proposed RGCA. The remaining configuration is unchanged. On MUUFL, RGCA improves OA, AA, and Kappa by 1.90%, 1.59%, and 2.35%, respectively. On Houston, the improvements are 1.03%, 1.16%, and 0.03%, respectively. These results show that residual gating improves cross-modal interaction over conventional cross-attention.

In Case 3, CTM is incorporated together with RGCA to further refine the modality-specific classification tokens through bi-directional information exchange. On MUUFL, this results in additional improvements of 2.46% in OA, 2.55% in AA, and 3.36% in Kappa over Case 2, demonstrating the effectiveness of token-level refinement after cross-modal alignment. On Houston, CTM further improves AA and Kappa from 93.97% to 94.23% and from 91.05% to 91.69%, respectively, while OA shows only a marginal decrease of 0.10%. This suggests that CTM primarily enhances balanced class-wise discrimination rather than uniformly improving all evaluation metrics.

Finally, Case 4 integrates SSTR with RGCA and CTM. SSTR further enhances the HSI representation by refining spectral–spatial information before cross-modal fusion. On MUUFL, the complete model achieves further gains of 1.68% in OA, 0.29% in AA, and 1.92% in Kappa over Case 3. On Houston, the inclusion of SSTR leads to improvements of 1.51% in OA, 0.84% in AA, and 1.96% in Kappa. Overall, compared with the baseline, the complete LGCF-Net improves OA, AA, and Kappa by 6.04%, 4.43%, and 7.63% on MUUFL, and by 2.44%, 2.26%, and 2.63% on Houston, respectively. These results demonstrate that the proposed modules provide complementary contributions and that their combined integration consistently produces the strongest overall performance across datasets with different characteristics.

A fine-grained ablation is further conducted to evaluate SPyA and MdSA within SSTR. RGCA and CTM are kept enabled in all settings. On MUUFL, removing SPyA reduces OA, AA, and Kappa from 91.19%, 92.41%, and 88.43% to 89.47%, 91.40%, and 86.18%, respectively. Removing MdSA reduces them to 88.36%, 90.90%, and 84.80%.

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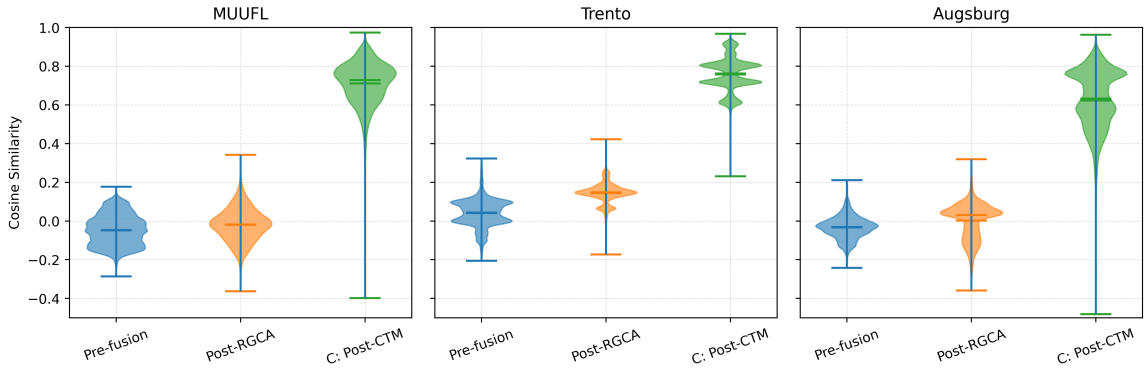


Fig. 12. Cosine similarity between HSI and LiDAR embeddings across progressive fusion stages for the MUUFL, Trento, and Augsburg datasets.

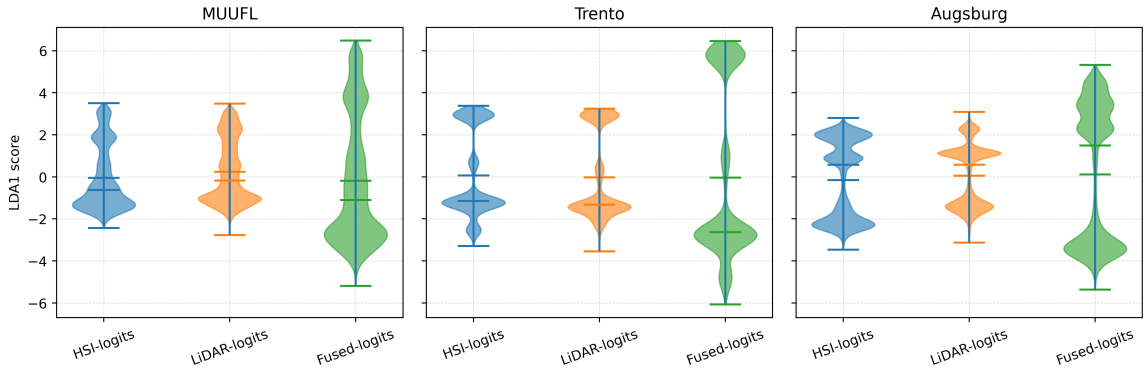


Fig. 13. LDA-based visualisation of decision-level representations for the MUUFL, Trento, and Augsburg datasets, comparing HSI-only, LiDAR-only, and fused branches.

A similar trend is observed on Houston. Without SPyA, the model achieves 92.07% OA, 92.90% AA, and 91.42% Kappa. Without MdSA, it achieves 92.75% OA, 93.10% AA, and 92.09% Kappa. The complete SSTR achieves the best performance on both datasets. These results support the complementary contributions of SPyA and MdSA to spectral–spatial refinement.

B. CMCT Parameter Analysis

Table VIII compares the parameter counts of different interaction mechanisms within CMCT while keeping the remaining network unchanged. The proposed CMCT contains 0.4446 M parameters, which is almost identical to the 0.4445 M parameters required by LGTE with standard cross-attention. This indicates that the residual gating mechanism introduces negligible additional parameter overhead. The convolutional, standard Transformer, and window-attention alternatives require fewer parameters, showing that the proposed CMCT achieves its local–global and gated cross-modal interaction with a moderate increase in model capacity.

C. Modality Alignment and Feature Discriminability Analysis

Statistical analysis is conducted to examine the effectiveness of modality fusion. To assess the effectiveness of cross-modal fusion, we analyze (i) the geometric alignment of modality-specific representations using cosine similarity, and (ii) class discriminability via supervised LDA projections.

TABLE VIII
PARAMETER COMPARISON OF DIFFERENT INTERACTION MECHANISMS WITHIN CMCT.

Interaction mechanism	Params (M)
Convolution + RGCA	0.3782
Standard Transformer + RGCA	0.2994
Window Self-Attention + RGCA	0.2994
LGTE + Standard Cross-Attention	0.4445
LGTE + RGCA (CMCT)	**0.4446**

First, cosine similarity is employed to quantify cross-modal alignment between HSI and LiDAR embeddings at three progressive fusion stages, as illustrated in Fig. 12. This analysis provides insight into how semantic agreement between the two modalities evolves throughout the fusion process. As expected, the pre-fusion representations exhibit the lowest similarity, indicating that HSI and LiDAR encode complementary but largely independent information prior to any interaction. Following the introduction of the Residual-Gated Cross-Attention (RGCA) module, a noticeable increase in cosine similarity is observed, indicating that RGCA facilitates effective cross-modal information exchange and encourages partial semantic alignment while still preserving modality-specific characteristics. A further substantial increase in similarity is then observed post-CTM, reflecting stronger consistency in the class-discriminative evidence produced by the two modalities. Importantly, this progressive alignment

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behaviour is consistently observed across all three datasets, demonstrating the robustness and general applicability of the proposed fusion strategy.

Subsequently, Linear Discriminant Analysis (LDA) is employed to assess the discriminative characteristics of the learned decision-level representations. Fig. 13 presents LDA-based visualisations for three datasets: MUUFL, Trento, and Augsburg. Each subplot corresponds to one dataset, while the three violins within each subplot represent the distributions of decision-level representations obtained from HSI-only, LiDAR-only, and fused branches, respectively. For all three datasets, the HSI and LiDAR representations exhibit relatively compact distributions with limited spread along the LDA axis, indicating restricted discriminative variation when each modality is considered independently. In contrast, the fused representations display a broader spread along the discriminant direction, reflecting increased between-class variance captured by LDA. This behaviour suggests that feature fusion enables a more discriminative representation space, providing representation-level evidence of enhanced class separability across all three datasets.

VI. Conclusion

In this paper, we proposed LGCF-Net, a unified multimodal framework for HSI–LiDAR land-cover classification that jointly models intra-modal local–global dependencies and inter-modal feature interactions. The proposed architecture integrates spectral–spatial feature refinement with convolution-embedded transformer encoding and residual-gated cross-modal attention, enabling effective exploitation of complementary spectral and structural information. Experimental evaluations on all four widely used HSI–LiDAR benchmark datasets demonstrate the effectiveness of the proposed framework. In particular, LGCF-Net achieves the best OA on MUUFL, Trento, and Augsburg, while attaining the highest AA and Kappa on Houston. It also improves the overall accuracy on the challenging MUUFL dataset by 1.84% compared with the strongest baseline. These results confirm that structured local–global feature modelling combined with adaptive cross-modal fusion can significantly enhance multimodal representation learning for land-cover classification.

Future work will investigate lightweight variants of the proposed architecture to improve computational efficiency. Further investigation will focus on label efficiency and generalization under limited supervision. The framework can also be extended to other multimodal remote sensing tasks, such as large-scale land-cover mapping and cross-sensor domain adaptation.

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